Practice workbook · Expanding a bracket with a negative or fractional power

Expanding (1 + kx)^n for rational n in ascending powers of x up to a stated term, with coefficients simplified.

18 question parts: 1 worked-example retry and 17 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-SER-18

1. Question 9 · part a · 3 marks

2020 JAN · Paper 01

Worked-example retry · Attempt part a only.

9 (a) Expand $\sqrt{1-4x}$ in ascending powers of x up to and including the term in x^3, giving each coefficient as an integer.

(3)

(b) Use your expansion with a suitable value for x to obtain an estimate of $\sqrt{0.76}$
Give your answer correct to 4 decimal places.

(3)

(c) Hence find, to 3 decimal places, an estimate of $\sqrt{19}$

(2)

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1. Question 9 · part a · 3 marks · Working and answer

Working and answer

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2. Question 9 · part a · 3 marks

2022 JAN · Paper 01

Further practice · Attempt part a only.

9 (a) Expand $\frac{1}{\sqrt{1-2x}}$ in ascending powers of x up to and including the term in x^3

Simplify each term as far as possible.

(3)

(b) Show that $\frac{1}{\sqrt{0.96}} = \frac{5\sqrt{6}}{12}$

Show all your working clearly.

(2)

(c) Express $\frac{1}{5\sqrt{6}-12}$ in the form $\frac{a\sqrt{6}}{c} + b$ where a, b and c are integers.

Show all your working clearly.

(2)

Using the expansion you found in part (a) with a suitable value of x

(d) find an estimate, to 5 decimal places, of $\frac{9}{5\sqrt{6}-12}$

(4)

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2. Question 9 · part a · 3 marks · Working and answer

Working and answer

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3. Question 6 · part b · 3 marks

2023 JAN · Paper 01

Further practice · Attempt part b only.

6 Given that $(8 + 3x)^{\frac{1}{3}}$ can be expressed in the form $p(1 + qx)^{\frac{1}{3}}$ where p and q are constants,

(a) find the value of p and the value of q

(2)

(b) Hence, expand $(8 + 3x)^{\frac{1}{3}}$ in ascending powers of x up to and including the term in x^2, expressing each coefficient as an exact fraction in its lowest terms.

(3)

Using the expansion found in part (b) with a suitable value of x

(c) show that $\sqrt[3]{9} \approx \frac{599}{288}$

(2)

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3. Question 6 · part b · 3 marks · Working and answer

Working and answer

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4. Question 6 · part b · 3 marks

2025 JUN · Paper 01

Further practice · Attempt part b only.

6 (a) Show that $\sqrt{4-x} = A\left(1-\frac{x}{B}\right)^{\frac{1}{2}}$ where A and B are integers to be found.

(2)

(b) Hence expand $\sqrt{4-x}$ in ascending powers of x up to and including the term in x^3
Give each coefficient as an exact fraction in its lowest terms.

(3)

(c) Use your expansion with a suitable value of x to obtain an estimate of $\frac{\sqrt{305}}{10}$

Give your answer correct to 5 decimal places.

(3)

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4. Question 6 · part b · 3 marks · Working and answer

Working and answer

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5. Question 6 · part b · 3 marks

2019 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

6 Given that $\sqrt{9-x}$ can be expressed in the form $p(1+qx)^{\frac{1}{2}}$ where p and q are constants

(a) find the value of p and the value of q.

(2)

(b) Hence expand $\sqrt{9-x}$ in ascending powers of x up to and including the term in x^3 expressing each coefficient as an exact fraction in its lowest terms.

(3)

Using the expansion you found in part (b) with a suitable value of x,

(c) find an estimate to 5 decimal places for the value of $\sqrt{\frac{31}{4}}$

(3)

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5. Question 6 · part b · 3 marks · Working and answer

Working and answer

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6. Question 1 · part a · 3 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part a only.

1 (a) Expand $\left(1+\frac{x}{4}\right)^8$ in ascending powers of x up to and including the term in x^3

Give each coefficient in its simplest terms.

(3)

(b) Use your expansion with a suitable value of x to obtain an approximation, to 4 decimal places, of $(1.035)^8$

(3)

(Total for Question 1 is 6 marks)

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6. Question 1 · part a · 3 marks · Working and answer

Working and answer

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7. Question 3 · part a · 3 marks

2020 NOV · Paper 02

Further practice · Attempt part a only.

3 (a) Expand

$$(1 + px)^{-5} \quad p \neq 0$$

in ascending powers of x, up to and including the term in x^4
Give each term in its simplest form.

(3)

The coefficient of x^r in the expansion is c_r

Given that $c_4 = 2c_3$

(b) find the value of p.

(2)

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7. Question 3 · part a · 3 marks · Working and answer

Working and answer

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8. Question 9 · part a · 3 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

9 (a) Expand $(1+2x)^{-\frac{1}{3}}$ in ascending powers of x up to and including the term in x^3 expressing each coefficient as a fraction in its lowest terms.

(3)

(b) Find the range of values of x for which your expansion is valid.

(1)

$$f(x) = \frac{2+kx^2}{(1+2x)^{\frac{1}{3}}}$$

(c) Obtain a series expansion of f (x) in ascending powers of x up to and including the term in x^3
Give your coefficients in terms of k where appropriate.

(3)

The coefficient of x^3 in the series expansion of f (x) is $-\frac{8}{3}$

(d) Find the exact value of k

(2)

(e) Hence, using algebraic integration, estimate the value of

$$\int_{0.1}^{0.2} f(x)\,dx$$

Give your answer to 4 decimal places.

(5)

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8. Question 9 · part a · 3 marks · Working and answer

Working and answer

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9. Question 2 · part a · 4 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

2 (a) Expand $\frac{2}{\sqrt{1+3x}}$ in ascending powers of x up to and including the term in x^3

Express each coefficient as a fraction in its simplest terms where appropriate.

(4)

(b) State the range of values of x for which the expansion is valid.

(1)

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9. Question 2 · part a · 4 marks · Working and answer

Working and answer

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10. Question 1 · part whole · 3 marks

2025 NOV · Paper 01

Further practice · Attempt part whole only.

1 Expand $\left(1-\frac{x}{4}\right)^{-2}$ in ascending powers of x up to and including the term in x^3

Where appropriate, express each coefficient as an exact fraction in its lowest terms.

(3)

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(Total for Question 1 is 3 marks)

10. Question 1 · part whole · 3 marks · Working and answer

Working and answer

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11. Question 5 · part b · 3 marks

2022 JUN · Paper 01

Further practice · Attempt part b only.

5 Given that $(2 + 3x)^{-1}$ can be expressed in the form $p(1 + qx)^{-1}$ where p and q are constants,

(a) find the value of p and the value of q

(2)

(b) Hence expand $(2 + 3x)^{-1}$ in ascending powers of x up to and including the term in x^3, expressing each coefficient as an exact fraction in its lowest terms.

(3)

$$f(x) = \frac{1 + x}{2 + 3x}$$

(c) Obtain a series expansion for $f(x)$, in ascending powers of x up to and including the term in x^3, expressing each coefficient as an exact fraction in its lowest terms.

(2)

(d) Hence use algebraic integration to obtain an estimate, to 4 decimal places, of

$$\int_0^{0.5} f(x)\,dx$$

(4)

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11. Question 5 · part b · 3 marks · Working and answer

Working and answer

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12. Question 7 · part a · 3 marks

2024 JUN · Paper 02

Further practice · Attempt part a only.

7 (a) Expand $\left(1+2x^2\right)^{-\frac{3}{4}}$ in ascending powers of x up to and including the term in x^6

Express each coefficient as an exact fraction in its lowest terms.

(3)

$$f(x) = \frac{(2+kx)}{\left(1+2x^2\right)^{\frac{3}{4}}} \quad \text{where } k \neq 0$$

(b) Obtain a series expansion for $f(x)$ in ascending powers of x up to and including the term in x^5

Give each coefficient in terms of k where appropriate.

(2)

The coefficient of the term in x^5 is fourteen times the coefficient of the term in x^2

(c) Find the value of k

(2)

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12. Question 7 · part a · 3 marks · Working and answer

Working and answer

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13. Question 9 · part b · 3 marks

2020 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

9 Given that $\frac{1}{(2-x)^3}$ can be written as $p(1-qx)^{-3}$

(a) find the value of p and the value of q.

(2)

(b) Expand $\frac{1}{(2-x)^3}$ in ascending powers of x up to and including the term in x^3 and express each coefficient as an exact fraction in its lowest terms.

(3)

$$f(x) = \frac{a+bx}{(2-x)^3} \text{ where } a \text{ and } b \text{ are integers}$$

The first three terms of the expansion of $f(x)$ are $\frac{3}{8} - \frac{43}{16}x + cx^2$

(c) Find the value of a and the value of b.

(3)

(d) Find the exact value of c.

(2)

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13. Question 9 · part b · 3 marks · Working and answer

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14. Question 9 · part b · 3 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

9 (a) Write $\frac{3}{(3-x)^3}$ in the form $a(1-bx)^{-3}$
where a and b are fractions in their lowest terms.

(2)

(b) Expand $\frac{3}{(3-x)^3}$ in ascending powers of x up to and including the term in x^3
Express each coefficient as a fraction in its lowest terms.

(3)

(c) (i) Use a suitable value of x with your expansion in part (b), to obtain an approximation for $\frac{24}{125}$ to 5 decimal places.

(ii) Find the percentage error, to 2 decimal places, of your approximation from the actual value.

(4)

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14. Question 9 · part b · 3 marks · Working and answer

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15. Question 10 · part b · 3 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

10 (a) Show that $\frac{3}{\sqrt{9-3x}} = \left(1-\frac{x}{3}\right)^{-\frac{1}{2}}$

(2)

(b) Hence expand $\frac{3}{\sqrt{9-3x}}$ in ascending powers of x up to and including the term in x^3 expressing each coefficient as an exact fraction in its lowest terms.

(3)

$$f(x) = \frac{1+2x}{\sqrt{9-3x}}$$

(c) Find the expansion of $3f(x)$ in ascending powers of x up to and including the term in x^3 expressing each coefficient as an exact fraction in its lowest terms.

(4)

(d) Hence, using algebraic integration, obtain an approximation to 6 significant figures for

$$\int_{0.1}^{0.2} \frac{1+2x}{\sqrt{9-3x}}\,dx$$

(4)

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15. Question 10 · part b · 3 marks · Working and answer

Working and answer

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16. Question 7 · part a · 3 marks

2023 JUN · Paper 02

Further practice · Attempt part a only.

7 (a) Expand $\left(1+\frac{x}{3}\right)^{-3}$ in ascending powers of x up to and including the term in x^3

Where appropriate express each coefficient as an exact fraction in its lowest terms.

(3)

(b) Write down the range of values of x for which your expression is valid.

(1)

(c) Express $(3+x)^{-3}$ in the form $P(1+Qx)^{-3}$ where P and Q are rational numbers whose values should be stated.

(2)

$$f(x)=\frac{(1+4x)}{(3+x)^3}$$

(d) Obtain a series expansion for $f(x)$ in ascending powers of x up to and including the term in x^2

(2)

(e) Hence, using algebraic integration, obtain an estimate of $\int_0^{0.2} f(x)\,dx$

Give your answer to 5 significant figures.

(3)

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16. Question 7 · part a · 3 marks · Working and answer

Working and answer

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17. Question 5 · part a · 3 marks

2020 NOV · Paper 01 (regional R)

Further practice · Attempt part a only.

5 (a) Expand $\sqrt{1-x}$ in ascending powers of x up to and including the term in x^3
Give each coefficient as an exact fraction in its lowest terms.

(3)

(b) Using your expansion with a suitable value of x, obtain an approximation, to 6 decimal places, of $\sqrt{0.92}$

(3)

(c) Hence find an approximation, to 5 decimal places, of $\sqrt{23}$

(2)

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17. Question 5 · part a · 3 marks · Working and answer

Working and answer

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18. Question 9 · part a · 3 marks

2023 NOV · Paper 01

Further practice · Attempt part a only.

9 (a) Expand $\left(1-8x^2\right)^{\frac{1}{2}}$ in ascending powers of x, up to and including the term in x^6 giving each coefficient as an integer.

(3)

$$g(x)=\frac{a+bx}{\sqrt{1-8x^2}} \quad \text{where } a \text{ and } b \text{ are prime numbers}$$

Given that the fourth and fifth terms, in ascending powers of x, in the series expansion of g(x) are $20x^3$ and $48x^4$ respectively,

(b) find the value of a and the value of b

(4)

Using the first five terms, in ascending powers of x, in the series expansion of g(x)

(c) obtain an estimate, to 4 significant figures, of $\int_0^{0.2} g(x)\,dx$

(4)

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18. Question 9 · part a · 3 marks · Working and answer

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